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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/223,425	05/30/2025	Yu Fang	SONA-0010-U01-C01-C10-C01	3984

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EXAMINER	
ALHARBI, ADAM MOHAMED	

ART UNIT	PAPER NUMBER
3663	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 19/223,425	Applicant(s) Fang et al.	
	Examiner ADAM ALHARBI	Art Unit 3663	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 05/30/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 05/30/2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of Claims

This Office Action is in response to the application filed on May 30, 2025. Claims 1-20 are presently pending and are presented for examination.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-20 are rejected under 35 U.S.C. 103 as being unpatentable over U.S. Pub. No. 20210375077 (hereinafter, "Kawahara") in view of U.S. Pub. No. 20200090428 (hereinafter, "Prozesky").

Regarding claim **1**, Kawahara discloses **an apparatus comprising:**

a vehicle event graphical user interface (GUI) ("The input unit 62 is constituted by a human-machine interface such as a touch panel, operation keys, and the like" (para 0027)) **configured to interpret a trigger description value from a user** ("the mechanic inputs the vehicle state (a fault state or the like), which is asked from the user, into the input unit 12 of the failure diagnosis device 10" (para 0046))**, the trigger description value including a trigger condition** ("A description concerning a procedure of a failure diagnosis process performed by the failure diagnosis device 10 will be presented with reference to FIGS. 7 and 8. Prior to executing the failure diagnosis process by the failure diagnosis device 10" (para 0046))**;**

a request circuit configured to determine a response action value in response to the trigger description value ("the identification unit 22 identifies the highest order DTC of the systems as being the true DTC. For example, in the case of the state shown in FIG. 9, the identification unit 22 identifies

the highest order DTC of the system 8" (para 0060))**, the response action value including at least one of a vehicle data identifier configured to identify vehicle data to be captured in response to the trigger condition** (" the identification unit 22 may identify, as the true trouble code, the trouble code that is first in the recording order from among the plurality of recorded trouble codes" (para 0076)) **or an alert execution description to be transmitted in response to the trigger condition** ("a case is assumed in which the DTCs of codes B, E, H, and I are input. As shown in FIG. 6, the DTC grouping of the codes B, E, H, and I includes the system 2 made up from codes B, E, and I, and the system 8 made up from code H. In the case that a failure (in this case, a chain failure) related to the system 2 is occurring in the vehicle 80, the meter 90 displays the warning lights 2 to 4 as shown in FIG. 11" (para 0058))**; and**

a cloud interface configured to receive ("based on the URL read out by the identification unit 22, the information calling unit 24 accesses the website provided by the server 102 via the communications interface 20, and an external network 100 such as a public line or the like" (para 0052)) **at least one of:**

at least a portion of the identified vehicle data ("the information calling unit 24 downloads the repair manual corresponding to the true DTC from the server 102, and outputs a display instruction to the display unit 18" (para 0052))**; or**

an alert response value determined in response to the alert execution description ("The measured values are then read and if need be supplied to a data converter/translator that can interpret the measured values as appropriate. On the basis of that, a warning can then be output or generated, for example" (para 0069))**.**

However, Kawahara does not explicitly teach

an alert response value determined in response to the alert execution description.

Prozesky, in the same field of endeavor, teaches

an alert response value determined in response to the alert execution description ("The measured values are then read and if need be supplied to a data converter/translator that can interpret the measured values as appropriate. On the basis of that, a warning can then be output or generated, for example" (para 0069))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Kawahara with the teachings of Prozesky in order to provide smartphone app can send the whole diagnosis command to the DataPlug 300 to read a specific piece of information; see Prozesky at least at [0065].

Regarding claim **2**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the GUI is configured to display a vehicle use case template of a plurality of vehicle use case templates in response to a template selection** ("The mechanic connects the connector 74 of the data collection device 60 to the data link connector 96 of the vehicle 80, and performs a predetermined operation on the input unit 62" (para 0046))**.**

Regarding claim **3**, Kawahara discloses **the apparatus of claim 2.** Additionally, Kawahara discloses **wherein the GUI is configured to display a portion of a plurality of vehicle use template identifiers** ("The computation unit 64 of the data collection device 60 collects the DTCs from each of the ECUs 82, and causes the DTCs to be displayed on the display unit 70. Next, the mechanic inputs the DTCs displayed on the display unit 70 of the data collection device 60" (para 0046))**, and wherein the GUI determines the portion … and a location value** ("The DTCs are used when repairing the vehicle. For example, a mechanic (a technician or the like) in charge of repairs connects a failure diagnosis device to the vehicle, and identifies the locations of the failures based on the DTCs displayed on a screen" (para 0003))**.**

However, Kawahara does not explicitly teach

…based on at least one of an authorization value…

Prozesky, in the same field of endeavor, teaches

…based on at least one of an authorization value… ("it is therefore further possible for authorization of a terminal 100 of the user 600 to be effected" (para 0049))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Kawahara with the teachings of Prozesky in order for legal data protection reasons; see Prozesky at least at [0049].

Regarding claim **4**, Kawahara discloses **the apparatus of claim 3.** Additionally, Kawahara discloses **wherein the location value corresponds to at least one of: location of the apparatus and a location of a vehicle** ("The DTCs are used when repairing the vehicle. For example, a mechanic (a technician or the like) in charge of repairs connects a failure diagnosis device to the vehicle, and identifies the locations of the failures based on the DTCs displayed on a screen" (para 0003))**.**

Regarding claim **5**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the GUI is configured to display a plurality of vehicle use template identifiers** ("The mechanic connects the connector 74 of the data collection device 60 to the data link connector 96 of the vehicle 80, and performs a predetermined operation on the input unit 62" (para 0046))**, and wherein the GUI is configured to indicate a portion of the plurality of vehicle use template identifiers are unavailable based on a vehicle data collection parameter value** ("in the case of a failure that does not involve illumination of a warning light 98, the warning light information 42 is left blank. The vehicle state information 44 is indicative of states (faults) that appear in the vehicle 80 when failures corresponding to the DTCs occur" (para 0042))**.**

Regarding claim **6**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the GUI is configured to display a vehicle use template including a plurality of vehicle data identifiers** ((Fig. 11, #30) and "an identification unit configured to identify a true trouble

code based on the system information, in a case that the plurality of trouble codes are recorded; and an output unit configured to output an identification result of the identification unit" (para 0009))**...**

However, Kawahara does not explicitly teach

...based on an authorization value.

Prozesky, in the same field of endeavor, teaches

...based on an authorization value ("it is therefore further possible for authorization of a terminal 100 of the user 600 to be effected" (para 0049))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Kawahara with the teachings of Prozesky in order for legal data protection reasons; see Prozesky at least at [0049].

Regarding claim **7**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the alert response value includes at least one of: an alert criterion, an alert type, an alert content, and an alert location** ((Fig. 11, #30) and "The meter ECU 82d controls the meter 90, and displays warning lights 98 in accordance with the failures" (para 0024))**.**

Regarding claim **8**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the request circuit is configured to reject the trigger description value in response to a vehicle data collection parameter value** (Fig. 7, #S4)**.**

Regarding claim **9**, Kawahara discloses **the apparatus of claim 8.** Additionally, Kawahara discloses **wherein the vehicle data collection parameter value indicates the identified vehicle data cannot be captured by a vehicle** (Fig. 7, #S4)**.**

Regarding claim **10**, Kawahara discloses **the apparatus of claim 1.** Additionally, Kawahara discloses **wherein the trigger description value includes a plurality of trigger conditions including the trigger condition** (Fig. 11, #30)**, and wherein the identified vehicle data is to be captured in response to the plurality of trigger conditions** (Fig. 11, #30)**.**

Regarding claim **11**, Kawahara discloses **the apparatus of claim 10.** Additionally, Kawahara discloses **wherein the plurality of trigger conditions use a plurality of vehicle data types** (Fig. 11, #30)**.**

Regarding claim **12**, Kawahara discloses **a method comprising:**

operating a user device including a vehicle graphical user interface ("The input unit 62 is constituted by a human-machine interface such as a touch panel, operation keys, and the like" (para 0027))**, a request circuit** ("the mechanic inputs the vehicle state (a fault state or the like), which is asked from the user, into the input unit 12 of the failure diagnosis device 10" (para 0046))**, and a cloud interface** ("based on the URL read out by the identification unit 22, the information calling unit 24 accesses the website provided by the server 102 via the communications interface 20, and an external network 100 such as a public line or the like" (para 0052))**;**

interpreting a trigger description value from a user (" the identification unit 22 may identify, as the true trouble code, the trouble code that is first in the recording order from among the plurality of recorded trouble codes" (para 0076))**, the trigger description value including a trigger condition** ("A description concerning a procedure of a failure diagnosis process performed by the failure diagnosis device 10 will be presented with reference to FIGS. 7 and 8. Prior to executing the failure diagnosis process by the failure diagnosis device 10" (para 0046))**;**

determining a response action value in response to the trigger description value ("the identification unit 22 identifies the highest order DTC of the systems as being the true DTC. For example, in the case of the state shown in FIG. 9, the identification unit 22 identifies the highest order DTC of the system 8" (para 0060))**, the response action value including at least one of a vehicle data identifier configured to identify vehicle data to be captured in response to the trigger condition** (" the identification unit 22 may identify, as the true trouble code, the trouble code that is first in the recording order from among the plurality of recorded trouble codes" (para 0076)) **or an alert execution description to be transmitted in response to the trigger condition** ("a case is assumed in which the

DTCs of codes B, E, H, and I are input. As shown in FIG. 6, the DTC grouping of the codes B, E, H, and I includes the system 2 made up from codes B, E, and I, and the system 8 made up from code H. In the case that a failure (in this case, a chain failure) related to the system 2 is occurring in the vehicle 80, the meter 90 displays the warning lights 2 to 4 as shown in FIG. 11" (para 0058))**; and**

receiving at least one of at least a portion of the identified vehicle data ("the information calling unit 24 downloads the repair manual corresponding to the true DTC from the server 102, and outputs a display instruction to the display unit 18" (para 0052))**.**

However, Kawahara does not explicitly teach

or an alert response value determined in response to the alert execution description.

Prozesky, in the same field of endeavor, teaches

or an alert response value determined in response to the alert execution description ("The measured values are then read and if need be supplied to a data converter/translator that can interpret the measured values as appropriate. On the basis of that, a warning can then be output or generated, for example" (para 0069))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Kawahara with the teachings of Prozesky in order to provide smartphone app can send the whole diagnosis command to the DataPlug 300 to read a specific piece of information; see Prozesky at least at [0065].

Regarding claim **13**, Kawahara discloses **the method of claim 12.** Additionally, Kawahara discloses **comprising:**

displaying a portion of a plurality of vehicle use template identifiers; and displaying a vehicle use case template of a plurality of vehicle use case templates in response to a template selection ("The mechanic connects the connector 74 of the data collection device 60 to the data link connector 96 of the vehicle 80, and performs a predetermined operation on the input unit 62" (para 0046))**.**

Regarding claim **14**, Kawahara discloses **the method of claim 12.** Additionally, Kawahara discloses **comprising:**

displaying a plurality of vehicle use template identifiers ("The computation unit 64 of the data collection device 60 collects the DTCs from each of the ECUs 82, and causes the DTCs to be displayed on the display unit 70. Next, the mechanic inputs the DTCs displayed on the display unit 70 of the data collection device 60" (para 0046))**; and**

indicating a portion of the plurality of vehicle use template identifiers are unavailable based on a vehicle data collection parameter value (Fig. 7, #S4)**, wherein the vehicle data collection parameter value indicates the identified vehicle data cannot be captured by a vehicle** (Fig. 7, #S4)**.**

Regarding claim **15**, Kawahara discloses **the method of claim 14.** Additionally, Kawahara discloses **wherein the data collection parameter value included a location of a vehicle** ("The DTCs are used when repairing the vehicle. For example, a mechanic (a technician or the like) in charge of repairs connects a failure diagnosis device to the vehicle, and identifies the locations of the failures based on the DTCs displayed on a screen" (para 0003))**.**

Regarding claim **16**, Kawahara discloses **the method of claim 12.** Additionally, Kawahara discloses **comprising:**

displaying a vehicle use case template including a plurality of vehicle data identifiers … ("The mechanic connects the connector 74 of the data collection device 60 to the data link connector 96 of the vehicle 80, and performs a predetermined operation on the input unit 62" (para 0046))**.**

However, Kawahara does not explicitly teach

…based on an authorization value.

Prozesky, in the same field of endeavor, teaches

…based on an authorization value ("it is therefore further possible for authorization of a terminal 100 of the user 600 to be effected" (para 0049))**.**

One of ordinary skill in the art, before the time of filing, would have been motivated to modify the disclosure of Kawahara with the teachings of Prozesky in order for legal data protection reasons; see Prozesky at least at [0049].

Regarding claim **17**, Kawahara discloses **the method of claim 12.** Additionally, Kawahara discloses **comprising:**

rejecting the trigger description value in response to a vehicle data collection parameter value (Fig. 7, #S4)**.**

Regarding claim **18**, Kawahara discloses **the method of claim 17.** Additionally, Kawahara discloses **wherein the vehicle data collection parameter value indicates the identified vehicle data cannot be captured by a vehicle** (Fig. 7, #S4)**.**

Regarding claim **19**, Kawahara discloses **the method of claim 12.** Additionally, Kawahara discloses **wherein the trigger description value includes a plurality of trigger conditions including the trigger condition** (Fig. 11, #30)**, and wherein the identified vehicle data is to be captured in response to the plurality of trigger conditions** (Fig. 11, #30)**.**

Regarding claim **20**, Kawahara discloses **the method of claim 19.** Additionally, Kawahara discloses **wherein the plurality of trigger conditions use a plurality of vehicle data types** (Fig. 11, #30)**.**

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ADAM ALHARBI whose telephone number is (313)446-6621. The examiner can normally be reached on M-F 11:00AM – 7:30PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abby Flynn can be reached on (571) 272-9855. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ADAM M ALHARBI/
Primary Examiner, Art Unit 3663